

Probing AlphaEarth Embeddings for Urban Heat

MUSA 650 Final Project

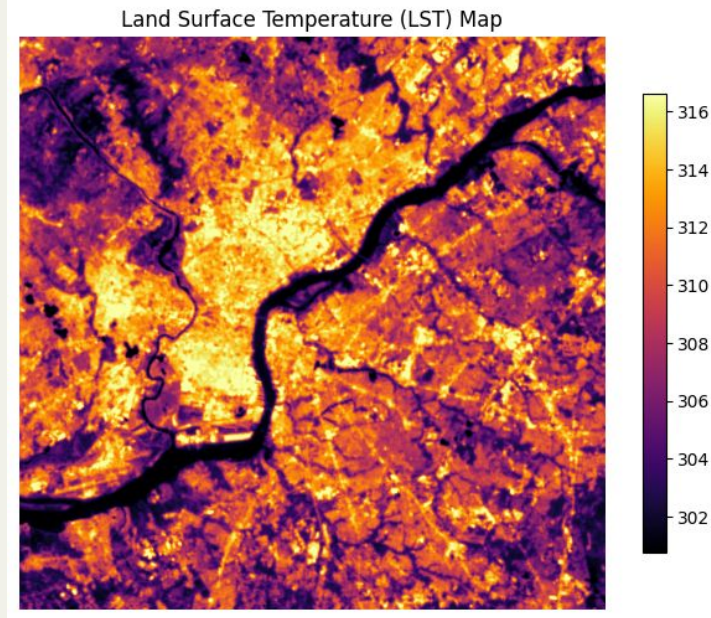
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Urban Heat is Spatially Uneven

Why it matters:

- Heat exposures varies sharply within cities, often at neighborhood and block scales
- Planners need interpretable maps to target cooling, greening, and emergency response
- A subjective vulnerability index can be useful descriptively, but is a weak supervised-learning target



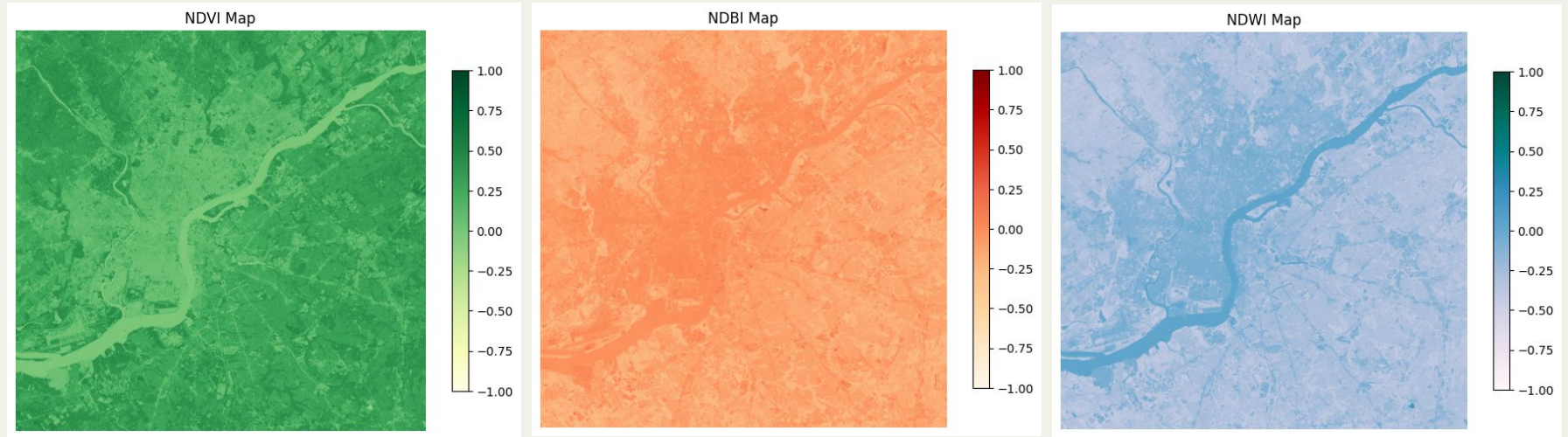
Approach

Predict continuous Landsat-derived LST from two feature sets:

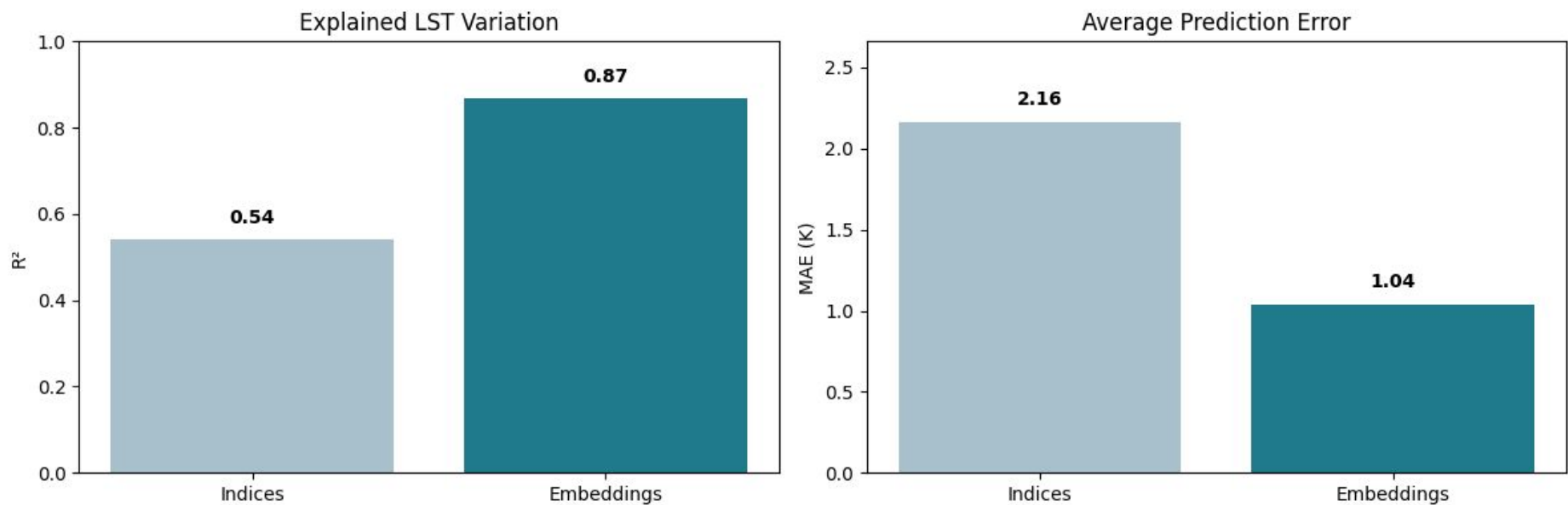
- **Baseline: NDVI + NDBI + NDWI**
- **Primary: 64 AlphaEarth embedding dimensions**



Data

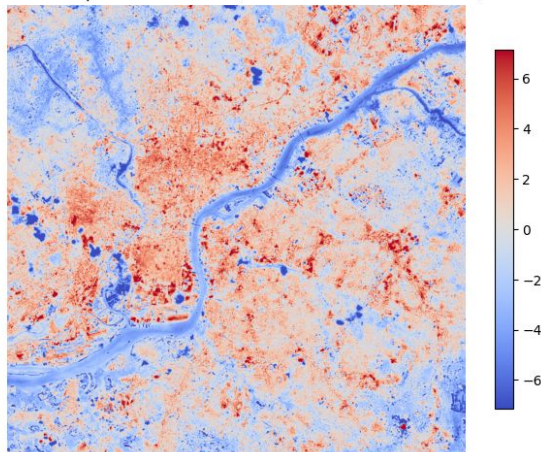


Key Result

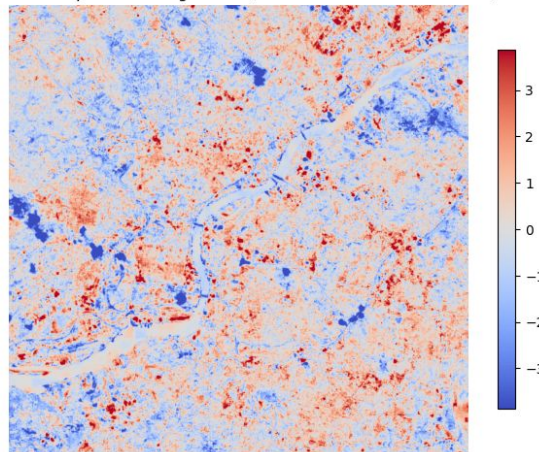


Residual Maps

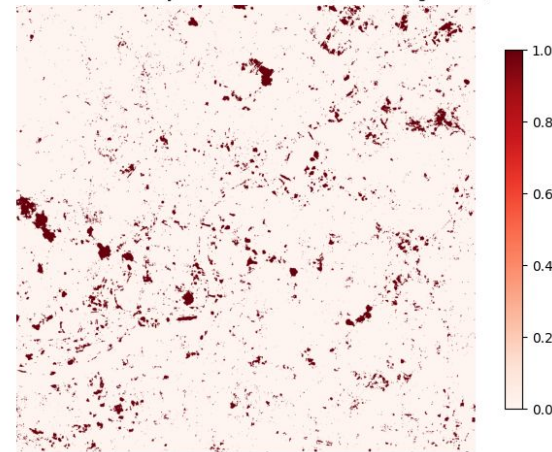
Residual Map: Baseline Model (Observed LST - Predicted LST)



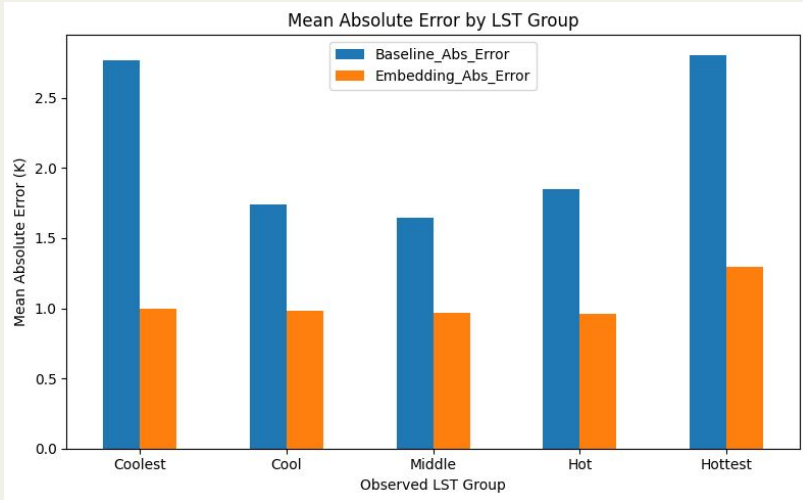
Residual Map: Embedding Model (Observed LST - Predicted LST)



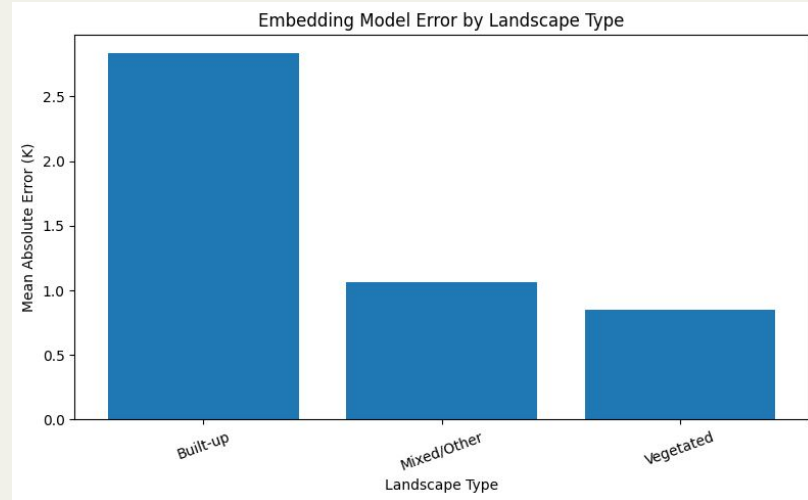
Worst 5% of Pixels by Absolute Residual (Embedding Model)



What the Model Gets Right & Wrong



Across LST groups, the embedding model has lower error than the baseline, including the hottest pixels. But the hottest group is still harder than moderate conditions.



Landscape-type analysis suggests built-up pixels may be harder to predict, but that category has a small sample count in the current threshold rule, so interpret cautiously.

Limitations

The model is useful, but not a magic:

- Possible thermal leakage
- Random pixel split
- Single scene/ one-city



Takeaways for Urban Heat Analysis

What we learned and where this goes next:

- **Embeddings add value**
- **Residuals are actionable**
- **Use with caution**

Future work:

- **Replace random split with block-based spatial validation**
- **Repeat across multiple summer dates and additional cities**
- **Compare against embeddings known to exclude thermal inputs**
- **Overlay outputs with tree canopy, cooling centers, transit stops, and demographic vulnerability**

